

Solution Requirements

Date	14 July 2026
Team ID	
Project Name	HDI Predictor (Human Development Index Predictor)
Team Size	1 (Individual Project)
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Prediction	The system shall predict an HDI score (0–1) from life expectancy, expected schooling, mean schooling, and GNI per capita.	High
2	Classification	The system shall classify the predicted HDI into one of four UNDP tiers: Very High, High, Medium, Low.	High
3	Input Validation	The system shall reject out-of-range values (e.g. negative life expectancy) with a clear error message.	High
4	History Logging	The system shall persist every prediction (inputs + result + timestamp) to a database.	High
5	History Management	The system shall allow filtering by tier, deleting a single record, and clearing all history.	Medium
6	Data Insights	The system shall display global HDI statistics and EDA charts on a Home dashboard.	Medium
7	Autofill Helper	The system shall optionally let a user load a real country's 2021 indicators into the Predict form.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Priority
1	Accuracy	The regression model shall achieve $R^2 \geq 0.90$ on held-out test data.	High
2	Usability	Predict form fields shall start blank (no pre-filled defaults) to force intentional user input.	High
3	Accessibility	Pages shall use semantic HTML, labeled inputs, visible focus states, and a skip-to-content link.	Medium
4	Performance	A prediction request shall return in under 1 second.	Medium
5	Scalability	Model and storage access shall be modularized (model_utils.py, db_utils.py) to allow swapping components later.	Medium
6	Responsiveness	The UI shall remain usable on mobile-width screens.	Low